Cloud Security Hands-on Report

Apply Through Hands-on Tasks | Cloud Security Week

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Objective. To apply cloud security skills in practice by exploring a cloud environment, analyzing a security incident, reviewing logs, identifying misconfigurations and summarizing the results in an assessment report. Screenshots from the practical are included as evidence under each task.

1. Explore Cloud Environments

The AWS Management Console was opened to understand the environment before checking its security.

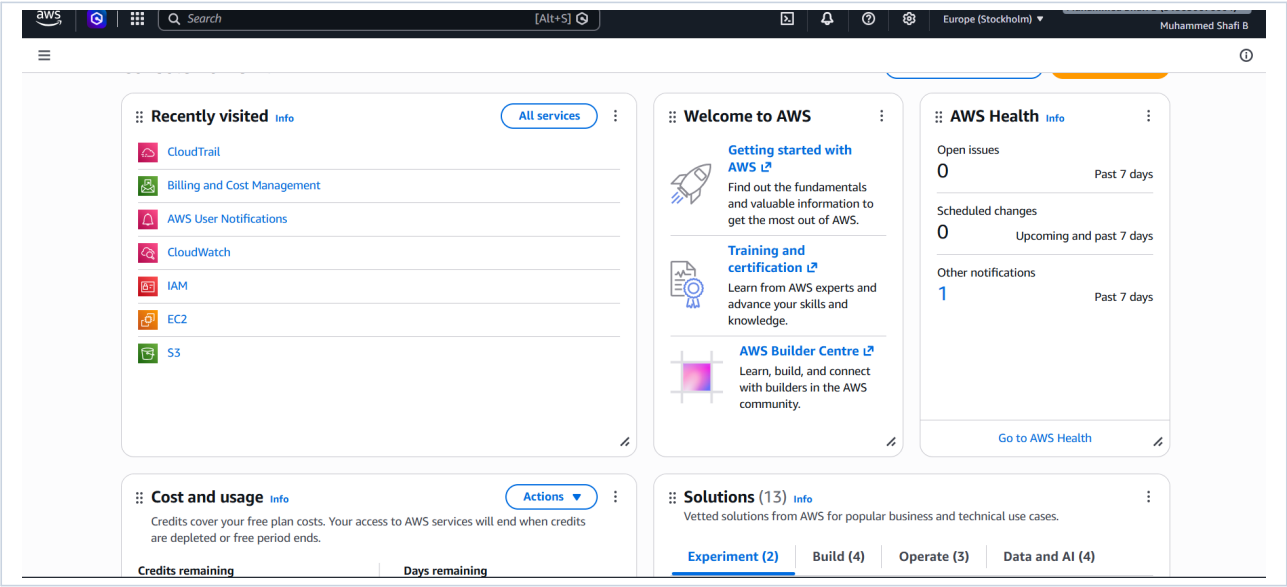


Figure 1: AWS Console home page (Europe - Stockholm region).

Observations

- Working region is Europe (Stockholm), eu-north-1.
- Recently used services: CloudTrail, Billing and Cost Management, User Notifications, CloudWatch, IAM, EC2 and S3.
- AWS Health shows 0 open issues in the past 7 days.
- Account is on a free plan; usage is covered by credits.

2. Analyze Cloud Security Incidents

A security monitoring console was used to analyze a suspicious activity report. The report lists access to a Windows registry key related to Remote Desktop (terminal server) settings.

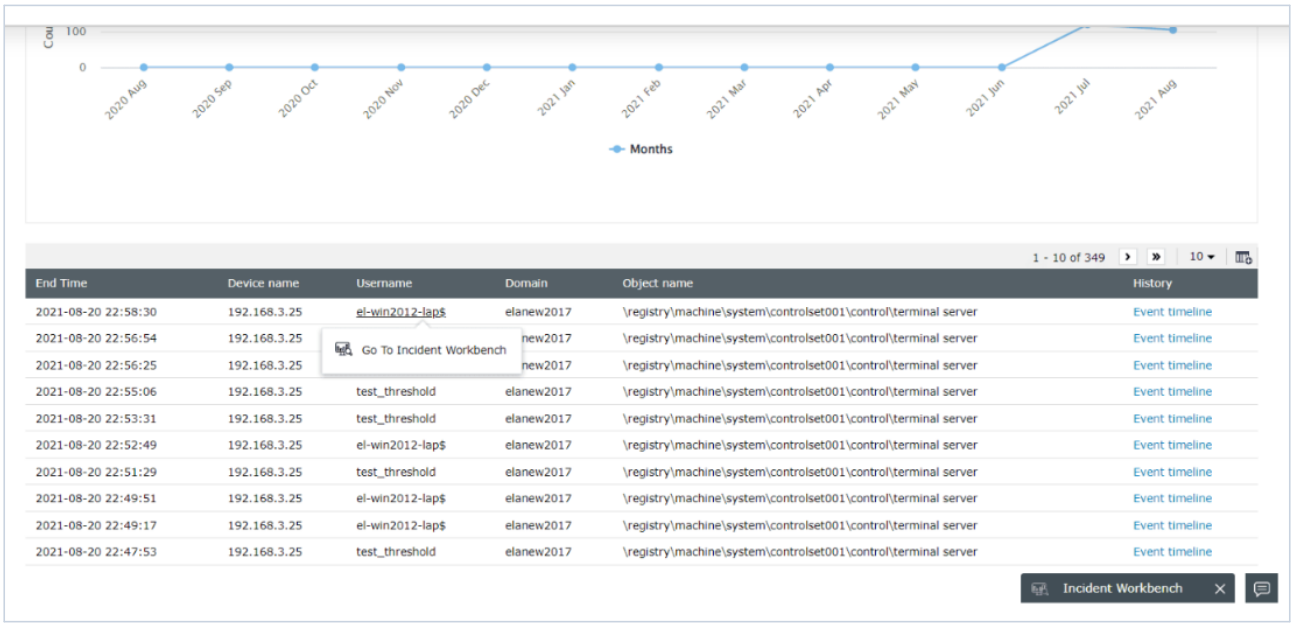


Figure 2: Event list and monthly trend with 349 matching events.

Observations

- The monthly chart is flat at zero from Aug 2020 to Jun 2021, then rises sharply in Jul and Aug 2021 (about 100 events per month).
- All visible events come from device 192.168.3.25 in domain elanew2017, for the registry path *\registry\machine\system\controlset001\control\terminal server*.
- Two accounts appear repeatedly: *el-win2012-lap$* (a machine account) and *test_threshold* (looks like a test account).
- Changes under this key can relate to Remote Desktop settings, so the activity should be checked against approved changes. The console offers an event timeline and Incident Workbench for further investigation.
- Suggested response: confirm whether the activity was planned testing; if not, restrict RDP access, review the two accounts and keep the event timeline as evidence.

3. Review Cloud Logs

AWS CloudTrail Event history was reviewed with the filter *Read-only = false* to show only events that change something or sign in.

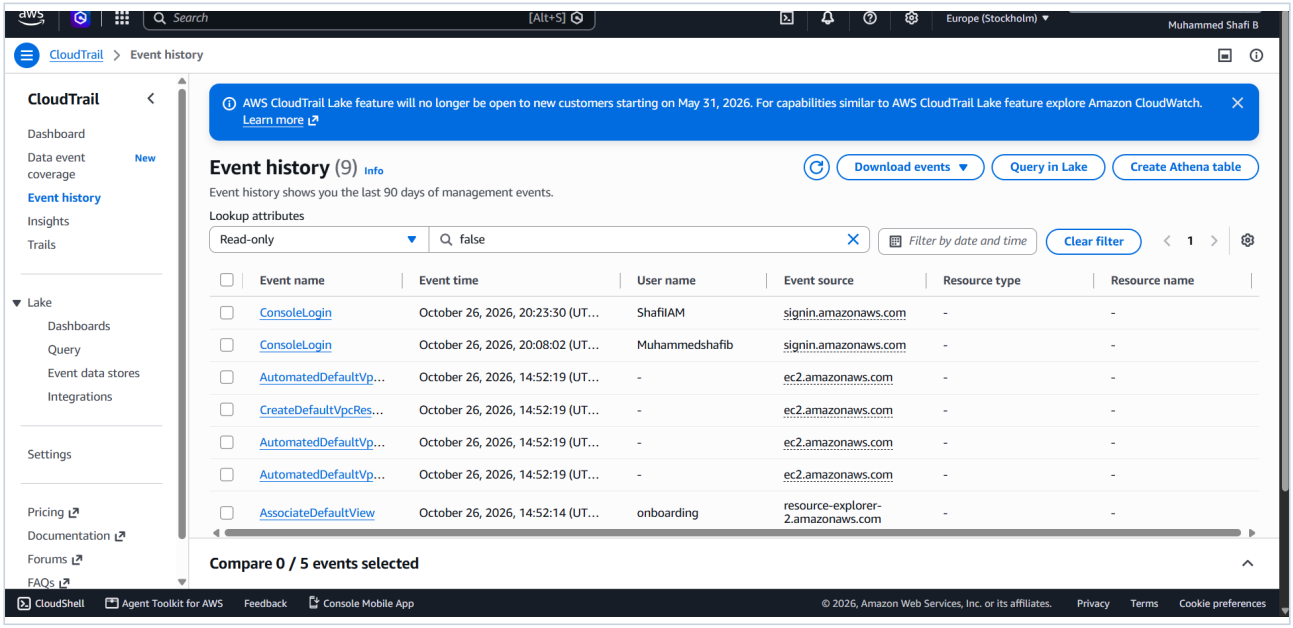


Figure 3: CloudTrail Event history (9 events, last 90 days).

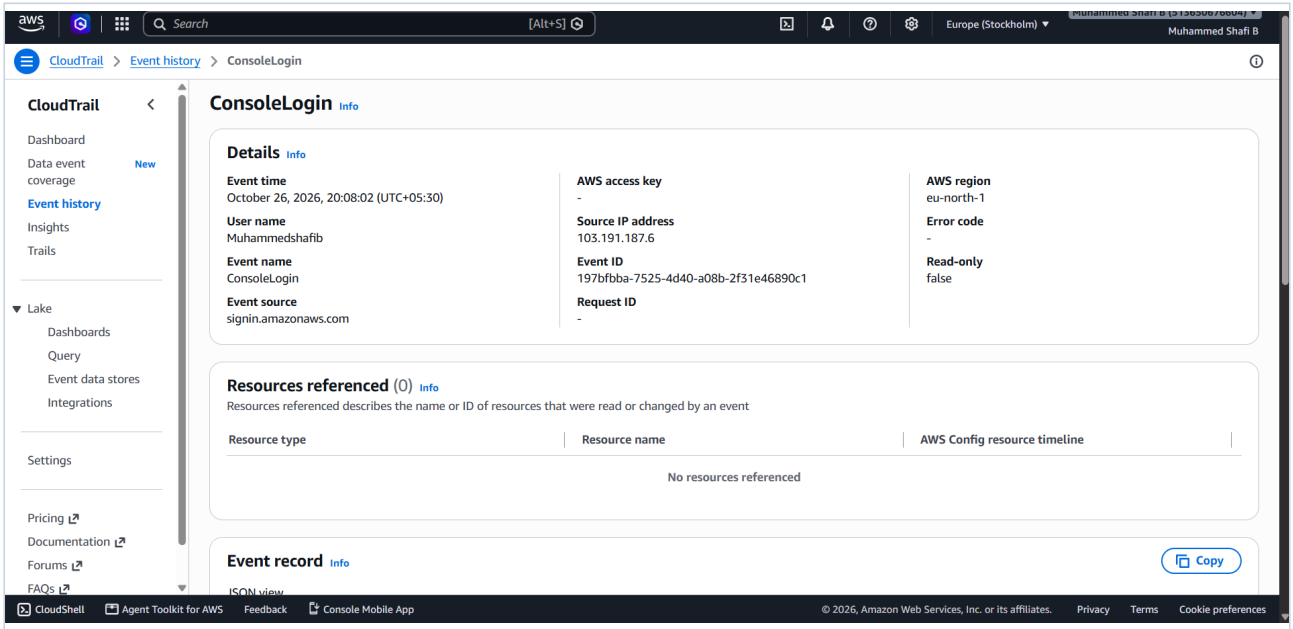


Figure 4: Details of a ConsoleLogin event.

Observations

- Two console sign-ins on 26 October 2026: user *Muhammedshafib* at 20:08:02 and *ShafiIAM* at 20:23:30 (times shown in UTC+05:30).
- Earlier events at about 14:52 are account setup activity (default VPC creation and onboarding).
- The detailed sign-in record shows source IP 103.191.187.6, region eu-north-1, no error code and no access key, so it was a normal console password login.
- No failed logins or unusual changes are visible in this list. The IP address should still be confirmed as the user's own network.

4. Identify Cloud Misconfigurations

The IAM user page was checked for weak identity and permission settings.

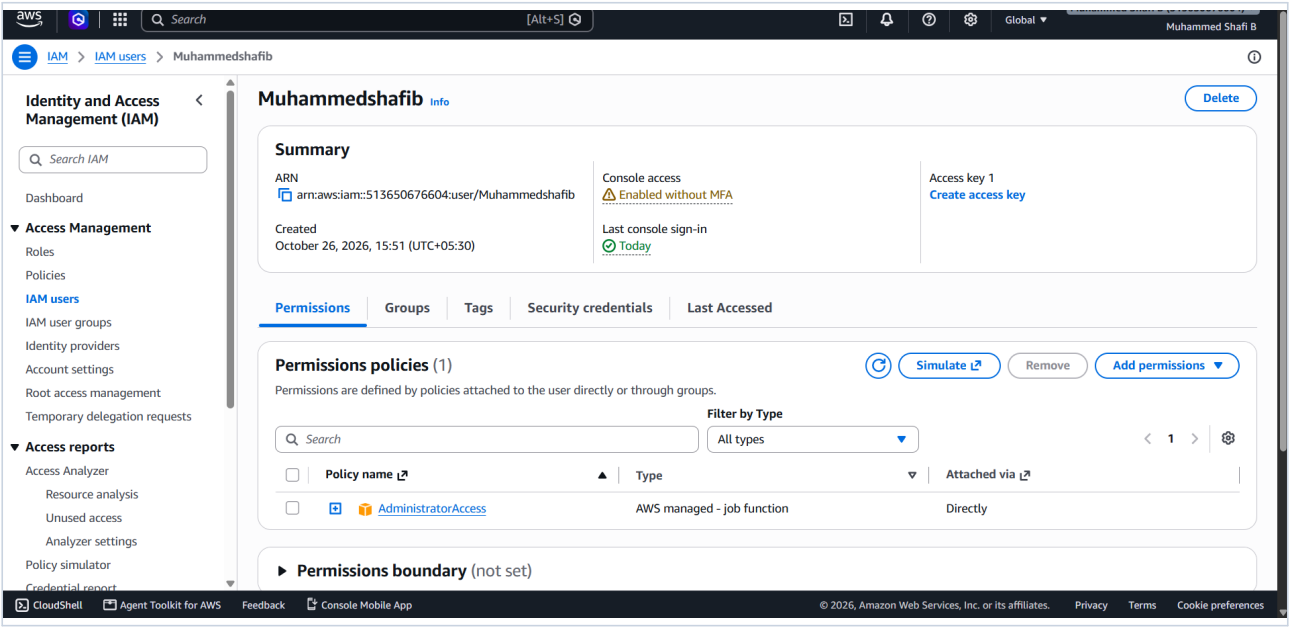


Figure 5: IAM user summary and permissions.

Findings

Finding	Evidence (Figure 5)	Risk	Recommendation
Console access without MFA	Summary shows "Enabled without MFA"	High	Enable MFA for this user and every console user
Full administrator rights	AdministratorAccess attached directly to the user	High	Use least privilege; use a group or role and only grant admin when needed
No permissions boundary	Permissions boundary shows "not set"	Medium	Set a boundary to cap the maximum permissions
Second console user	ShafiIAM also signed in (Figure 3)	Low	Confirm the account is needed and apply the same controls

Positive point: no access key has been created for this user, which reduces the risk of leaked keys.

5. Cloud Security Assessment Summary

Scope. One AWS account (eu-north-1) and one security monitoring report. Review was limited to the screenshots collected in this practical.

Overall result. No active attack is visible in the AWS logs, but the account has high-risk identity settings that should be fixed first. The monitoring report shows an unexplained rise in Remote Desktop-related registry activity that needs verification.

Priority actions

Priority	Action	Task
1 - Immediate	Enable MFA on all IAM users with console access	Task 4
2 - Immediate	Replace direct AdministratorAccess with least-privilege permissions	Task 4
3 - Short term	Verify the registry/RDP activity from 192.168.3.25 and review the two accounts	Task 2
4 - Short term	Confirm sign-in IP 103.191.187.6 and review the second IAM user	Task 3
5 - Ongoing	Keep CloudTrail enabled and review logs regularly; set up alerts	Task 3

Conclusion

The practical covered the full cloud security workflow: exploring the environment, analyzing an incident, reviewing logs, finding misconfigurations and reporting. Enabling MFA and reducing administrator permissions are the most important improvements.

Note: screenshots show an AWS account ID and IP addresses. Hide these before sharing the report outside your course.